Ch 5 · NIV and high-flow nasal oxygen

HFNO physiology

- Flow up to 60 L/min, FiO2 to 1.0, dead-space washout, low PEEP, humidification

Evidence

- FLORALI: HFNO ↓ 90-day mortality vs NIV/standard O2 in de novo hypoxaemia
- NIV first-line in COPD hypercapnia and cardiogenic pulmonary oedema; helmet NIV promising
- HFNO for pre-oxygenation and after extubation in low-risk patients

Failure

- ROX index (SpO2/FiO2 ÷ RR) < 3.85 at 12 h predicts failure; HACOR for NIV
- Tachypnoea, paradox, rising FiO2, shock, falling GCS → intubate; avoid delay

Key point: Delayed intubation after HFNO/NIV failure increases mortality.

Ch 21 · ARDS

Definition (Berlin)

- Within 1 week, bilateral opacities, not cardiac; PaO2/FiO2 on PEEP ≥ 5: 200–300 / 100–200 / ≤ 100
- 2023 global definition adds HFNO, SpO2/FiO2 and ultrasound

Ventilation

- Vt 6 mL/kg PBW, plateau ≤ 30, driving pressure < 15; higher PEEP if moderate–severe
- Permissive hypercapnia (pH > 7.2), SpO2 88–95%; avoid aggressive recruitment (ART)

Adjuncts

- Prone ≥ 16 h if P/F < 150 (PROSEVA); conservative fluids (FACTT)
- NMB for dyssynchrony (ROSE: no routine benefit); VV-ECMO for refractory cases (EOLIA)

Key point: Small tidal volumes and prone positioning are the two mortality-reducing interventions.

Ch 22 · Acute exacerbation of COPD

Initial treatment

- Controlled O2 to 88–92%; bronchodilators; prednisolone 40 mg 5 days; antibiotics if purulent

NIV

- pH < 7.35 and PaCO2 > 6.5 kPa: NIV ↓ intubation and mortality
- IPAP 15→20–25, EPAP 4–5; ABG at 1–2 h
- Contraindications: arrest, cannot protect airway, vomiting, facial trauma, undrained pneumothorax

Invasive ventilation

- Low RR, long expiratory time, measure PEEPi, external PEEP ≤ 80% PEEPi
- Target pre-morbid PaCO2/HCO3; early extubation to NIV

Key point: Over-oxygenation worsens hypercapnia – titrate to 88–92%.

Ch 23 · Acute severe asthma

Severity

- Life-threatening: PEF < 33%, SpO2 < 92%, normal PaCO2, silent chest, exhaustion, arrhythmia
- Near-fatal: raised PaCO2 or needing ventilation

Medical therapy

- Continuous salbutamol, ipratropium, steroids, IV magnesium 2 g
- Consider IV salbutamol/aminophylline/adrenaline, ketamine; heliox limited

Ventilation

- Ketamine induction, large ETT, fluid preload
- RR 8–10, Vt 6–8 mL/kg, I:E ≥ 1:4, plateau < 30; permissive hypercapnia
- Arrest/hypotension: disconnect and compress chest; exclude pneumothorax; ECMO rescue

Key point: Normal PaCO2 in a breathless asthmatic is a danger sign.

Ch 25 · VTE in the ICU

PE risk classes

- High risk: persistent hypotension or arrest; intermediate-high: RV dysfunction + troponin
- Low risk: sPESI 0, no RV strain

High-risk PE

- UFH + systemic thrombolysis (alteplase 100 mg/2 h) unless contraindicated
- Contraindicated: catheter-directed therapy or surgical embolectomy; VA-ECMO
- RV support: noradrenaline, dobutamine, avoid big fluid loads and intubation where possible

Prophylaxis

- LMWH > UFH (PROTECT); IPC if bleeding risk; adding IPC gives no extra benefit (PREVENT)

Key point: Thrombolysis is for haemodynamic instability – not for RV strain alone.

Ch 26 · Massive haemoptysis

Key facts

- Any volume threatening airway or gas exchange; > 90% from bronchial arteries
- Causes: bronchiectasis, TB/aspergilloma, cancer, CF, AVM, vasculitis, coagulopathy

Airway first

- Bleeding side down; large ETT (≥ 8.0); lung isolation (selective intubation, DLT, blocker)
- TXA, correct coagulopathy; bronchoscopy for localisation and local measures

Definitive

- CT angiography → bronchial artery embolisation (first-line)
- Surgery for localised disease after failed embolisation

Key point: Patients die of asphyxia, not blood loss – protect the good lung.

Ch 27 · Sedation and delirium

Sedation principles

- Analgesia first; RASS 0 to −2; daily awakening + breathing trial
- Propofol or dexmedetomidine over benzodiazepines; ABCDEF bundle

Delirium

- CAM-ICU or ICDSC; hypoactive commonest, worst prognosis
- Risk: age, dementia, alcohol, benzodiazepines, sepsis, restraints, sleep loss

Management

- Treat causes; reorientation, sleep, mobilisation, family, glasses/hearing aids
- Antipsychotics do not shorten delirium (MIND-USA, AID-ICU); dexmedetomidine for agitation

Key point: Delirium duration predicts long-term cognitive impairment.

Ch 28 · Prolonged mechanical ventilation

Definition and causes

- ≥ 21 days for ≥ 6 h/day
- ICU-acquired weakness, diaphragm dysfunction, overload, cardiac, malnutrition, delirium

Weaning

- Daily trach-collar trials of increasing length beat PSV wean (Jubran 2013)
- Inspiratory muscle training, mobilisation, nutrition, fluid removal, speaking valve

Tracheostomy and outcome

- TracMan: early vs late – no mortality difference
- About half liberated; 1-year mortality ≈ 50% – discuss goals

Key point: Prolonged ventilation is a multidisciplinary rehabilitation problem.

Ch 30 · Acute exacerbation of IPF

Definition

- Worsening < 1 month with new bilateral ground-glass not due to cardiac failure/fluid
- Exclude infection, PE, pneumothorax, drug toxicity, aspiration

Management

- Oxygen/HFNO, antibiotics until infection excluded, steroids (weak evidence)
- Cyclophosphamide harmful (EXAFIP); continue antifibrotic; palliate dyspnoea

Ventilation

- Invasive ventilation mortality > 80–90%
- Reserve for transplant bridge or reversible cause; early goals-of-care talk

Key point: Decide the ceiling of care before the patient needs intubation.

Ch 31 · Weaning from mechanical ventilation

Readiness

- Cause improving, P/F > 150–200 on PEEP ≤ 8, FiO2 ≤ 0.4–0.5, stable, awake, cough

SBT

- 30–120 min low PSV/T-piece; PSV 8 for 30 min helps success
- RSBI < 105; failure: RR > 35, SpO2 < 90%, HR/BP change, distress
- Pair with awakening trial (ABC)

After extubation

- High risk (age > 65, cardiac/lung disease, weak cough): prophylactic NIV ± HFNO (HIGH-WEAN)
- Low risk: HFNO; cuff leak test + steroids if stridor risk

Key point: Most patients pass their first SBT – test daily.

Ch 33 · Diffuse alveolar haemorrhage

Causes

- Capillaritis: ANCA vasculitis, anti-GBM, SLE; bland: coagulopathy, mitral stenosis; DAD: HSCT

Diagnosis

- Falling Hb, infiltrates, progressively bloody BAL, haemosiderin-laden macrophages > 20%
- ANCA, anti-GBM, ANA/dsDNA, urinalysis (red cell casts), renal biopsy

Treatment

- Methylprednisolone 0.5–1 g/day × 3; cyclophosphamide or rituximab
- Plasma exchange for anti-GBM (PEXIVAS: no benefit in ANCA overall)
- Supportive ventilation, RRT, ECMO

Key point: Haemoptysis may be absent – suspect DAH when Hb falls with new infiltrates.

Ch 34 · Pleural disease

Exudate vs transudate

- Light's criteria: protein ratio > 0.5, LDH ratio > 0.6, LDH > 2/3 ULN

Pleural infection

- pH < 7.2, low glucose, high LDH, septations = complicated; drain + antibiotics
- Small-bore drain adequate; tPA + DNase (MIST-2); VATS if failing

Pneumothorax on ventilator

- Rising pressures, hypoxia, hypotension, absent sliding on ultrasound
- Tension: immediate decompression then drain; persistent leak = bronchopleural fistula

Key point: Any pneumothorax on positive pressure needs a drain.

Ch 85 · Thoracic trauma

Life threats

- Airway obstruction, tension/open pneumothorax, massive haemothorax, flail chest, tamponade

Flail and contusion

- Regional analgesia (epidural, serratus/erector spinae block), HFNO/NIV, avoid overload
- Rib fixation ↓ ventilator days and pneumonia

Haemothorax

- Thoracotomy if > 1500 mL initially or > 200 mL/h for 2–4 h, or instability
- Retained haemothorax: VATS within 3–7 days

Key point: Analgesia is the main treatment that prevents respiratory failure after rib fractures.

Ch 90 · ECMO and ECCO2R

Indications (VV-ECMO)

- EOLIA: P/F < 50 for > 3 h, < 80 for > 6 h, or pH < 7.25 with $PaCO_2 \geq 8$ kPa for > 6 h
- Reversible disease, ventilation < 7 days

VV vs VA

- VV: lung support only; VA: heart + lung support
- VA problems: limb ischaemia, LV distension, differential hypoxia

Management and evidence

- Ultraprotective ventilation; blood flow → O_2, sweep → CO_2; anticoagulation, haemolysis watch
- CESAR and EOLIA support referral; ECCO2R (REST) no benefit

Key point: ECMO buys time for lungs to recover – it is not a treatment of the disease.